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Management Information Systems

(ICT 2242)

Proposal for a Sales and Inventory Management Information System for a Multi-Branch Supermarket

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Name: Tharinda Gimhana

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1. Introduction and Business Background

The following report was written from the viewpoint of an information systems consultant for a large retail chain that has multiple locations in Sri Lanka. The company offers grocery items, beverages, household goods, personal care items, clothes, and electronics among other items. Both in-store purchases and online orders for home delivery are available. At present, the firm maintains records in a manual system and individual files at all branches, creating many difficulties.

2. Current Business Processes and Existing Problems

Currently, each branch operates independently using physical books and individual Excel sheets. Due to this manual process, the supermarket faces the following daily challenges:

- **Inefficient Stock Monitoring:** The checkout counters are not linked to the store room. Counting the shelves manually is the only way of checking the stock levels.
- **Late Reorders:** There are no automatic notifications system. Therefore, staff only realize that the fast-moving items are out of stock once the shelves are empty. So the orders may get delayed.
- **Lack of Coordination between branches:** The head office finds it difficult to get information on stock levels in the branches without contacting them or getting weekly sheets.
- **Online and Walk-in Orders Conflict:** The online orders are made without taking into account the live stock levels in the branches. The order made online might have been purchased by a walk-in customer.
- **Poor Decision-Making:** It takes days for managers to gather sales data from branches manually for decision making.

3. Proposed Information System

These issues can be addressed by a centrally-located cloud-based Sales & Inventory Management Information System. The system links all supermarket branches, checkout counters, storerooms, online shopping portal, and suppliers to a single database such that every time a sale is made, products are delivered to the storeroom, or orders are made online, the inventory gets updated instantly.

4. System Users and Information Requirements

Different users in the supermarket needs different kinds of information from the system:

User Group	Main Role	Their Requirements
Cashiers and Counter Staff	Customer billing and payments	Item prices, barcodes, active discounts, and loyalty points.
Store and Warehouse Clerks	Stock handling and packing online orders	Current stock levels, expiry dates, reorder limits, and supplier deliveries.
Branch Managers	Managing daily branch work	Daily branch sales summaries, fast/slow-moving items, and low-stock alerts.
Top Management	Long term business planning	Monthly revenue reports, branch profit comparisons, and seasonal sales trends.
Customers and Suppliers	Buying goods or supplying stock	Live online stock status, order tracking, and clear digital Purchase Orders.

5. Data and System Requirements

- **Core Data Requirement:** Product data (product code, barcode, name, type, buying and selling price, expire date), quantity of the product present in each branch, sales bill details, customer details, and Purchase Orders and delivery date from supplier.
- **Hardware Requirement:** Point of Sale (POS) computers, barcode readers, receipt printer, and UPS backup power supply on counters. Handheld barcode reader is also used for checking stock of products in store room and a cloud based database will host the central database.

- **Software and Network Requirement:** Central relational database (MySQL), Point of Sale Billing System, Web Interface for the management and Head Office. Online shopping website is connected to the same database and each branch must have proper internet access.
- **Functional Requirements:** The system should real time update the stock status after each sale, delivery of the supplier and online orders. Automatic calculation of bills, automatic discount, loyalty point, low stock alert, and generation of reports every day and every month.
- **Integration and User Interface Requirement:** The system should integrate with the online payment gateway, SMS for customer offers, Purchase Orders of suppliers. Screens must be simple and designed differently according to cashier, store clerks and managers.

6. Support for Operations and Management

6.1 Transaction Processing System (TPS)

The POS billing counters and the checkout are the TPS systems. Scanning of the barcodes allows the calculation of the bill, printing the bill, making the entry of payment and reducing the stock at that branch's location.

6.2 Management Information System (MIS) and Decision Making

The MIS gathers daily data from the TPS and generates reports for the managers. The branch managers will receive the sales summary report on a daily basis and the products nearing their expiry date, which they will use for running discount offers on those products. The top management will receive the chart of the branch-wise profits and also the sales trends during festivals for budget planning.

6.3 Inventory and SCM

Each product will have the minimum order quantity defined in the system. If the stock goes below this quantity in any of the branches, it will send the low stock notification automatically along with the purchase order digitally to the vendor. Also, the batch expiry dates will be taken into account (First-Expired, First-out).

6.4 Customer Relationship Management (CRM)

This site links to the live stock of the closest branch. The moment an order is made via the Internet, these are reserved instantly so that there is no conflict between walk-in and Internet orders. Additionally, customers can register their mobile number to avail themselves of loyalty points and SMS discounts.

7. Security and Privacy Controls

- **Access Control Based on Roles**: Cashiers have access to billing screens only, store employees have access to updating stock, and only managers have access to financial reports and setting prices.
- **Encryption and Regular Backups:** Passwords of the users and online transactions are all encrypted. Automatic cloud backup occurs daily to avoid any loss of information due to any computer crash in one of our branches.
- **Keeping Audit Trail:** All bill cancellations and any updates in stock will be logged along with the identification number of staff member and date and time to prevent internal abuse.
- **Customer Information Security:** Customer's phone number and address details will be used solely for fulfilling orders, providing loyalty points, and offering discounts. No other person will get this information unless customer gives consent for it according to PDPA of Sri Lanka.
- **Secure Login and Inactivity Logout Feature:** All users will have a unique username with strong password, and the user account will be disabled as soon as that employee leaves the organization. Users will be logged out automatically when inactive for a period of time.

8. Problems in Implementation and Possible Solution

- **Opposition from staff members:** Staff members who are accustomed to manual books can resist the system. Solution: Provide simple and practical training in Sinhala and English prior to implementation.
- **High Implementation Cost:** Purchase of POS machine and scanners for each branch can be costly. Solution: Implement the system gradually, beginning with the main branch.
- **Hardness of migration:** Manual input of thousands of items in the database may lead to inaccuracies. Solution: Verify the physical inventory of the branch on a weekend.
- **Loss of Internet Connection:** Internet connection loss at the branch cannot disrupt billing process. Solution: Incorporate offline POS system which will record the bill offline.

9. Business Value and Competitive Advantage

The supermarket gains a number of advantages with regard to everyday activities and can distinguish itself among other supermarkets.

- **Savings of Time and Money:** By using barcodes and automatic bill calculation, waiting for bills is reduced. The staff will have enough time to assist the clients because there will be no need to manually calculate the products on the shelves.
- **Mitigation of Losses:** With the use of expiry notifications and FIFO, the supermarket sells products that are about to expire. Accurate stock accounting helps to avoid overstocking and shortage of stock.
- **Higher Customer Loyalty:** Through live inventory checking and order reservation, the supermarket delivers the right orders to the customers. Loyalty points and SMS offers stimulate customers to shop at the supermarket.
- **More Informed Management:** With sales dashboard, management receives up-to-date information regarding sales, profit, and stock. This allows to make quicker decisions than management of rival supermarkets who do not have automated accounting.

- **Better Control and Development:** Head office manages all branches through one system, so the management can easily control expenses, prevent fraud, and open new branches in the future.

10. Conclusion

There is a need to replace record books manually maintained by the salesmen with the Sales and Inventory Management Information System. This is because through the centralization of POS billing, branch inventories, online ordering, purchasing from suppliers, and management reporting, the organization will be able to function effectively.

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